STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM

SECMS — Salesforce Administrator Project

Complete Project Documentation

Technology Track: Salesforce Administrator

Platform: Salesforce CRM / Lightning Experience

Prepared as a structured implementation document covering data model, security, automation, validation, reports, dashboards, testing and deployment.

Source & Document Note

The supplied PDF identifies the Salesforce Administrator project as “Student Enrollment & Course Management System (SECMS)” and provides a Google Docs project-document link and a reference-video link. □filecite□turn0file0□L2-L4□

The linked Google Docs page could not be retrieved in this environment. Therefore, this PDF is a complete, ready-to-use SECMS project document built around the project title and Salesforce Administrator scope from the supplied reference, while clearly avoiding a claim that it is a verbatim copy of the inaccessible Google Doc.

If you upload/export the Google Doc itself as PDF/DOCX, this document can be revised to reproduce its exact headings, wording, screenshots, field names and step order.

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1. Project Overview

The Student Enrollment & Course Management System (SECMS) is a Salesforce-based application intended to centralize student records, course information and enrollment transactions. The system provides administrators with a structured way to create and maintain academic records and gives users a controlled view of the information relevant to their responsibilities.

The solution is designed as a Salesforce Administrator project, emphasizing declarative configuration such as custom objects, fields, relationships, validation rules, Flow automation, permissions, reports and dashboards.

High-level process: Student Registration → Course Creation → Course Availability → Enrollment → Status Management → Academic Tracking → Reporting.

Key stakeholders include administrators, academic staff, faculty/course coordinators and students. Access is controlled according to the user's role and required responsibilities.

2. Problem Statement

- Student information may be maintained in disconnected spreadsheets or manual records.
- Course information can become difficult to maintain when changes are made in multiple places.
- Enrollment transactions require consistent student-course relationships.
- Duplicate or incomplete records can reduce data quality.
- Administrators need consolidated reports for enrollment and course activity.
- Manual status updates can cause delays and inconsistent information.

3. Objectives

- Create a centralized Salesforce application for student and course management.
- Maintain accurate student profiles and course records.
- Track each student's enrollment in courses.
- Automate routine updates using Salesforce Flow.
- Apply validation rules to prevent invalid data.
- Provide role-based access to academic information.
- Create reports and dashboards for management visibility.
- Provide an auditable and scalable declarative solution.

4. Scope

In Scope

- Student master data
- Course master data
- Enrollment transaction management
- Enrollment status and dates
- Course capacity / availability tracking
- Basic faculty or course coordinator assignment
- Validation and automation
- Reports and dashboards
- Salesforce security configuration

Out of Scope / Optional Extensions

- Online payment gateway
- External university ERP integration
- Full learning-management content delivery
- Biometric attendance hardware integration
- Production email/SMS provider integration

5. Functional Requirements

ID	Requirement	Description
FR-01	Student Management	Create, edit, view and maintain student records.
FR-02	Course Management	Create and maintain courses, credits, capacity and status.
FR-03	Enrollment	Create an enrollment connecting a student and a course.
FR-04	Status	Track enrollment as Pending, Active, Completed, Dropped or Waitlisted.
FR-05	Validation	Prevent invalid dates, duplicates and missing mandatory data.
FR-06	Automation	Use Flow to update records and support notifications/status changes.
FR-07	Security	Restrict access based on profile/permission requirements.
FR-08	Reporting	Provide operational reports for students, courses and enrollments.

6. Salesforce Data Model

A practical SECMS data model can be implemented with the following custom objects:

Object	Purpose	Key Relationship
Student__c	Stores student master information.	Parent to Enrollment__c
Course__c	Stores course/catalog information.	Parent to Enrollment__c
Enrollment__c	Transactional record connecting student and course.	Lookup/Master-Detail to Student and Course
Faculty__c	Stores faculty/course coordinator information.	Related to Course
Course_Schedule__c	Optional schedule details for a course.	Related to Course and Faculty

Recommended architecture: use Enrollment__c as the junction/transaction object between Student__c and Course__c. This supports the many-to-many relationship between students and courses.

7. Objects and Fields

7.1 Student__c

Field	Type	Purpose
Student ID	Auto Number / Text	Unique student identifier.
First Name	Text	Student first name.
Last Name	Text	Student last name.
Email	Email	Student email address.
Phone	Phone	Contact number.

Field	Type	Purpose
Date of Birth	Date	Student date of birth.
Gender	Picklist	Student gender, if required by institution.
Address	Long Text Area	Residential/contact address.
Admission Date	Date	Date of admission.
Status	Picklist	Active, Inactive, Graduated, etc.

7.2 Course__c

Field	Type	Purpose
Course ID	Auto Number / Text	Unique course identifier.
Course Name	Text	Name of the course.
Course Code	Text	Unique academic code.
Description	Long Text Area	Course description.
Credits	Number	Credit value.
Capacity	Number	Maximum seats.
Available Seats	Number / Formula	Remaining capacity.
Start Date	Date	Course start date.
End Date	Date	Course end date.
Status	Picklist	Planned, Open, Closed, Completed.
Faculty	Lookup	Assigned faculty/coordinator.

7.3 Enrollment__c

Field	Type	Purpose
Enrollment ID	Auto Number	Unique enrollment number.
Student	Master-Detail / Lookup	Student being enrolled.
Course	Master-Detail / Lookup	Selected course.
Enrollment Date	Date	Date of enrollment.
Status	Picklist	Pending, Active, Completed, Dropped, Waitlisted.
Grade	Picklist/Text	Final academic result, when applicable.
Completion Date	Date	Date the course was completed.
Remarks	Long Text Area	Administrative notes.

7.4 Faculty__c and 7.5 Course_Schedule__c

Object	Suggested Fields
Faculty__c	Faculty ID, First Name, Last Name, Email, Phone, Department, Status
Course_Schedule__c	Course, Faculty, Day of Week, Start Time, End Time, Classroom, Schedule Status

8. Relationships

- Student__c → Enrollment__c: One student can have many enrollment records.
- Course__c → Enrollment__c: One course can have many enrollment records.
- Student__c ↔ Course__c: Many-to-many relationship implemented through Enrollment__c.
- Faculty__c → Course__c: A faculty member can coordinate or teach one or more courses.
- Course__c → Course_Schedule__c: A course can have multiple schedule records.
- Faculty__c → Course_Schedule__c: A faculty member can appear in multiple schedule records.

Relationship design should prioritize data integrity and reporting. The exact choice between Lookup and Master-Detail should follow the institution's sharing and ownership requirements.

9. Security and Access

Recommended access model:

Role/Profile	Typical Access
System Administrator	Full configuration and data access.
Academic Administrator	Create/update students, courses and enrollments; view reports.
Faculty / Coordinator	View assigned courses and relevant enrollment/academic information; update permitted fields.
Student User	View own student information and own enrollments, subject to portal/security design.

Security layers to configure:

- Profiles for baseline object permissions.
- Permission Sets for additional access without changing the base profile.
- Field-Level Security for sensitive fields.
- Organization-Wide Defaults and sharing rules where required.
- Role hierarchy for internal visibility.

10. Validation Rules

Rule	Logic / Business Purpose
Course End Date	End Date must not be earlier than Start Date.
Enrollment Date	Enrollment Date should not be later than the course start date when the institution requires pre-sta
Grade	Grade should be entered only when enrollment status is Completed.
Completion Date	Completion Date should be required when status is Completed.
Capacity	Course capacity must be greater than zero.
Course Code	Course Code must be unique through a unique field or duplicate-management approach.
Required Student Data	First Name, Last Name and Email should be present before activation where required.

Example Salesforce validation formulas (adapt field API names to the final configuration):

End date check:
End_Date__c < Start_Date__c

Completion date check:
AND(ISPICKVAL(Status__c, "Completed"), ISBLANK(Completion_Date__c))

Grade check:
AND(NOT(ISPICKVAL(Status__c, "Completed")), NOT(ISBLANK(Grade__c)))

11. Automation and Flows

Flow 1 — Enrollment Creation

- Trigger when an Enrollment__c record is created.
- Confirm the related course is open for enrollment.
- Set Enrollment Status to Active or Pending according to the business rule.
- Update course seat information where the design uses a maintained seat count.
- Create a notification/task when administrative review is required.

Flow 2 — Enrollment Completion

- When Status changes to Completed, require/confirm Completion Date.
- Allow grade entry and preserve the academic record.
- Update course/enrollment reporting data.

Flow 3 — Course Capacity

- When an enrollment becomes Active, increment the enrolled count or recalculate it.
- When an enrollment is Dropped, decrement the count or recalculate it.
- When capacity is reached, mark the course Closed or prevent further enrollment according to the selected rule.

Flow 4 — Student Status

- Support administrative status changes such as Active, Inactive or Graduated.

12. User Interface and Lightning App

- Create a Lightning App named SECMS.
- Navigation items: Students, Courses, Enrollments, Faculty, Schedules, Reports and Dashboards.
- Configure Lightning Record Pages for Student, Course and Enrollment.
- Use related lists so users can see enrollments from a Student and students from a Course.
- Add useful quick actions such as New Student, New Course and New Enrollment.
- Use Dynamic Forms where appropriate for conditional field visibility.

13. Reports and Dashboards

Report	Purpose
Students by Status	View active/inactive/graduated student counts.
Courses by Status	View open, closed and completed courses.
Enrollments by Course	Analyze the number of students enrolled in each course.
Enrollments by Student	Review each student's course registrations.
Enrollment Status Summary	Monitor active, completed, dropped and waitlisted records.
Course Capacity Report	Compare course capacity with enrollment count.
Faculty Course Assignment	Show courses and schedules assigned to faculty.

Suggested dashboard components:

- Total Students — metric
- Active Courses — metric
- Total Active Enrollments — metric
- Enrollments by Course — bar chart
- Enrollment Status — donut/pie chart
- Course Capacity Utilization — chart
- Recent Enrollment Activity — table

14. Sample Data

Sample Student Records

Student ID	Name	Email	Status
STU-001	Arun Kumar	arun@example.com	Active
STU-002	Priya S	priya@example.com	Active
STU-003	Rahul M	rahul@example.com	Active
STU-004	Divya R	divya@example.com	Graduated

Sample Course Records

Course Code	Course Name	Credits	Capacity	Status
CS101	Introduction to Computing	4	40	Open
DB201	Database Management	4	35	Open
SF301	Salesforce Administration	3	30	Open
WD202	Web Development	4	40	Planned

Sample Enrollment Records

Enrollment	Student	Course	Date	Status
ENR-001	STU-001	SF301	2026-09-01	Active
ENR-002	STU-002	DB201	2026-09-02	Active
ENR-003	STU-003	CS101	2026-09-03	Pending
ENR-004	STU-004	CS101	2026-01-10	Completed

15. Test Cases

TC	Scenario	Expected Result
TC01	Create valid student	Student record is saved successfully.
TC02	Create student without mandatory name	Validation prevents save.
TC03	Create valid course	Course is created with required details.
TC04	Course end date before start date	Validation error is displayed.
TC05	Create enrollment for open course	Enrollment is created.

TC	Scenario	Expected Result
TC06	Enroll in closed/full course	Enrollment is blocked or waitlisted according to rule.
TC07	Complete enrollment without completion date	Validation prevents save.
TC08	Enter grade for non-completed enrollment	Validation prevents invalid grade entry.
TC09	Dashboard refresh	Updated enrollment metrics are displayed.
TC10	Unauthorized user access	User cannot access restricted records/actions.

16. Implementation Procedure

Phase 1 — Create Salesforce App

- Open Setup → App Manager.
- Create a Lightning App named SECMS.
- Configure navigation items and branding.

Phase 2 — Create Custom Objects

- Create Student__c, Course__c, Enrollment__c, Faculty__c and optional Course_Schedule__c.
- Enable required options such as reports, activities and search as appropriate.

Phase 3 — Create Fields

- Create fields according to the data dictionary in Section 7.
- Mark required fields at the appropriate level.

Phase 4 — Configure Relationships

- Create Student and Course relationships on Enrollment.
- Add Faculty relationship to Course.
- Add related lists to parent record pages.

Phase 5 — Security

- Configure profiles, permission sets and field-level security.
- Set sharing defaults and rules based on organizational requirements.

Phase 6 — Validation

- Create and activate the validation rules in Section 10.

Phase 7 — Automation

- Build record-triggered flows for enrollment, capacity and completion processes.
- Test each flow independently with positive and negative scenarios.

Phase 8 — Reports and Dashboard

- Create the reports in Section 13.
- Create the SECMS dashboard and add components.

Phase 9 — Testing

- Load sample data.
- Execute all test cases.

- Fix validation, automation or access issues.

17. Deployment Checklist

Area	Checklist
Objects	All custom objects created and labeled correctly.
Fields	Required fields, data types and picklists verified.
Relationships	Student/Course/Enrollment relationships tested.
Security	Profiles, permission sets and FLS tested with representative users.
Validation	All rules activated and tested.
Flows	Flows activated, debugged and tested for recursion/edge cases.
UI	Lightning App and record pages assigned to intended users.
Reports	Reports return correct records and filters.
Dashboard	Dashboard components display expected values.
Data	Sample/initial data imported and checked for duplicates.

18. Expected Outcomes

- Centralized student and course records.
- Consistent enrollment tracking.
- Reduced manual data entry through automation.
- Improved data quality through validation.
- Controlled access to academic information.
- Improved visibility through reports and dashboards.
- A scalable declarative Salesforce foundation for future features.

19. Conclusion

The SECMS project demonstrates how Salesforce Administrator capabilities can be used to build a structured academic management application without requiring a large amount of custom code. The core design centers on Student, Course and Enrollment records, supported by relationships, validation, Flow automation, security and reporting.

The solution can be extended later with attendance, examinations, fees, notifications, student portals, external integrations and advanced analytics while retaining the same core data model.

20. Viva / Interview Questions

Question	Short Answer
Why use Salesforce for SECMS?	It provides configurable objects, security, automation, reporting and dashbo
Why is Enrollment a separate object?	It represents the transaction connecting students and courses and supports
Lookup vs Master-Detail?	Lookup provides a looser relationship; Master-Detail creates stronger depen
What is a validation rule?	A declarative rule that prevents a record from being saved when specified c
What is Flow?	Salesforce's declarative automation tool for record-triggered and other busi

Question	Short Answer
Why use Permission Sets?	To grant additional permissions without creating many custom profiles.
What is a Lightning Record Page?	A configurable page layout that controls the user experience for a record.
What reports are useful?	Enrollment by course, enrollment status, student status, course capacity and
How can the system be extended?	Attendance, payments, exams, notifications, portals and integrations can be

Appendix A — Suggested Salesforce Naming Convention

- Student object API name: Student__c
- Course object API name: Course__c
- Enrollment object API name: Enrollment__c
- Faculty object API name: Faculty__c
- Course Schedule object API name: Course_Schedule__c

Appendix B — Suggested Picklist Values

Student Status: Active, Inactive, Graduated
Course Status: Planned, Open, Closed, Completed
Enrollment Status: Pending, Active, Completed, Dropped, Waitlisted

End of Document

Implementation Evidence

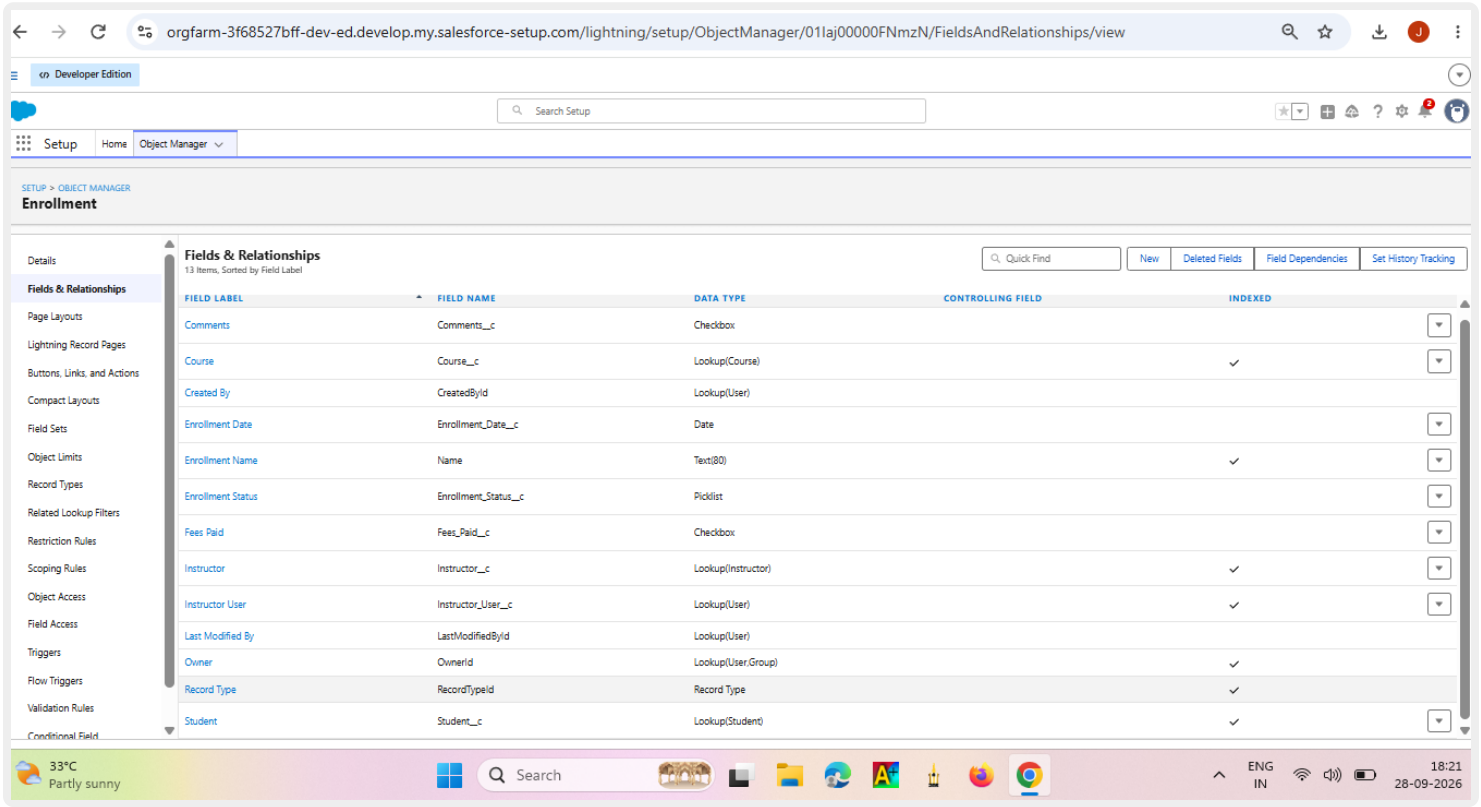
SECMS — Salesforce Administrator Project

The following pages contain the screenshots supplied for the completed Salesforce configuration. They document objects, fields, layouts, validation, flows, approval and reporting.

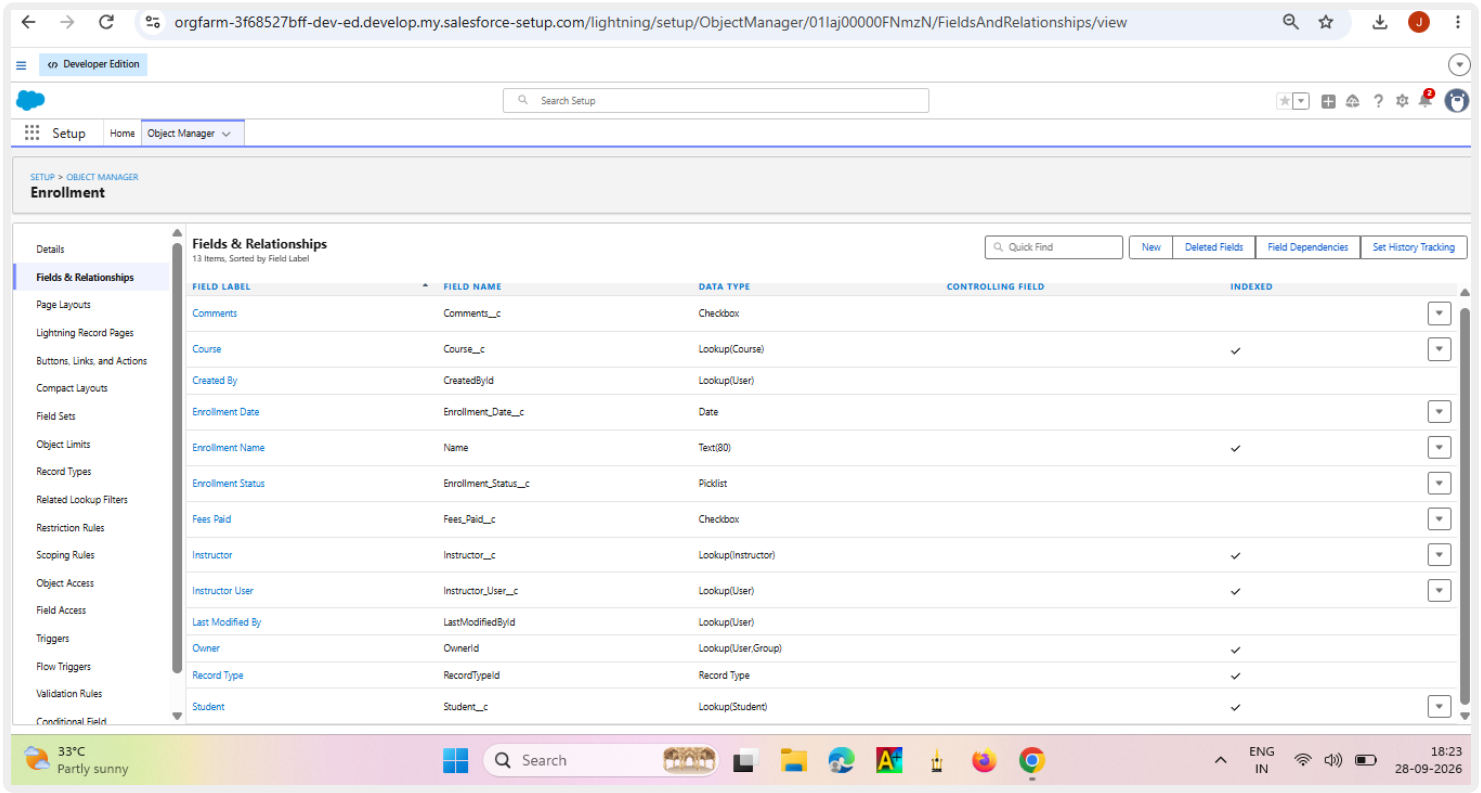
Screenshot Evidence Index

01. Enrollment Object — Fields & Relationships
02. Enrollment Object — Fields & Relationships (continued)
03. Enrollment Object — Page Layouts
04. Enrollment Validation Rule — Fees Must Be Paid Before Approval
05. Enrollment Object — Record Types
06. Student Object — Fields & Relationships
07. Student Object — Page Layouts
08. Student Validation Rule — Student Email Domain Validation
09. Course Object — Fields & Relationships
10. Course Object — Page Layouts
11. Instructor Object — Fields & Relationships
12. Instructor Object — Page Layouts
13. Flow — Auto Assign Instructor
14. Flow — Create Follow-up Task
15. Flow — Enrollment Approved Action
16. Approval Processes — Enrollment
17. Report — Pending Enrollment Approvals
18. Report — Revenue Report / Fees Paid
19. Report — Enrollments by Course
20. Report — Students by Status

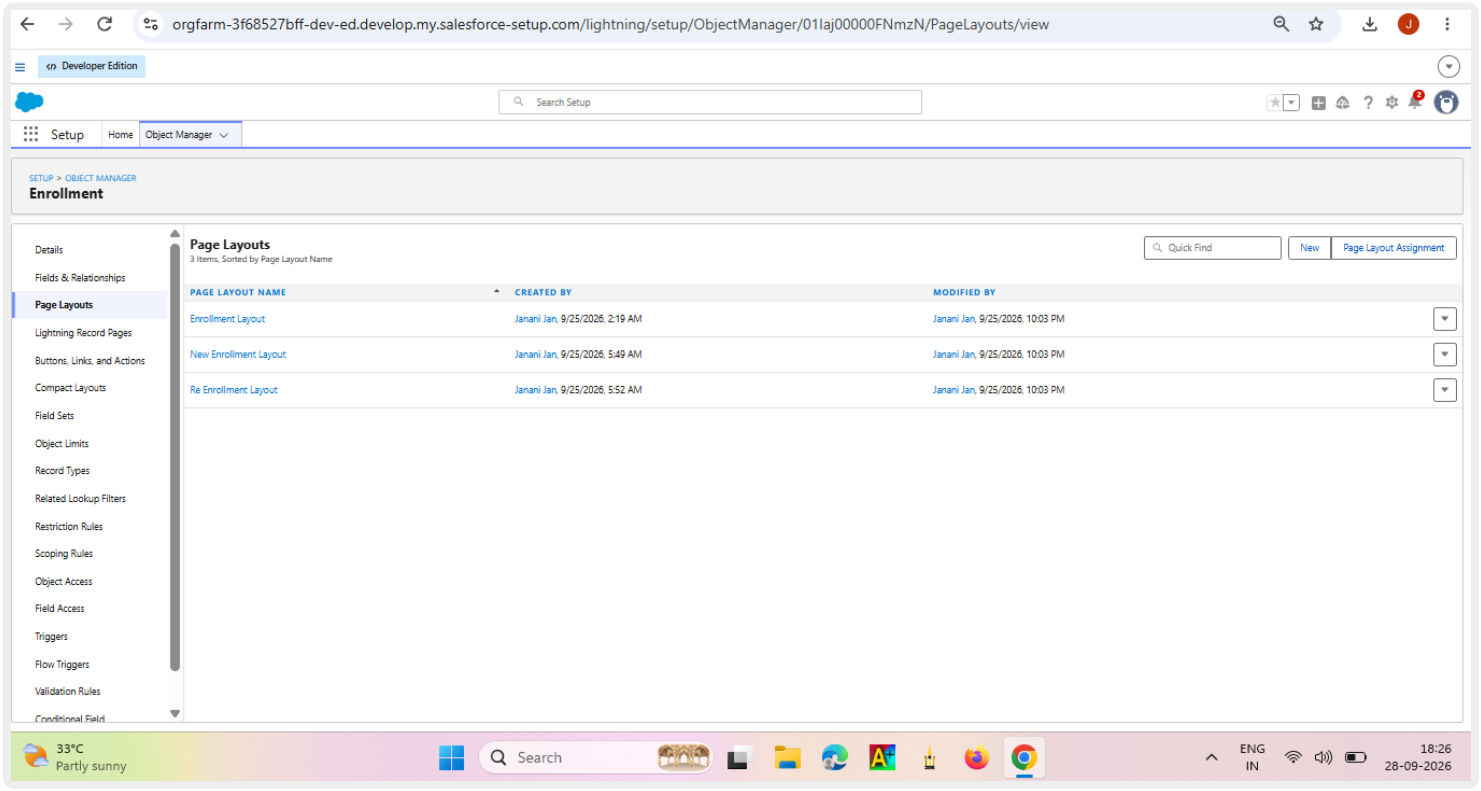
Screenshot 01 — Enrollment Object — Fields & Relationships



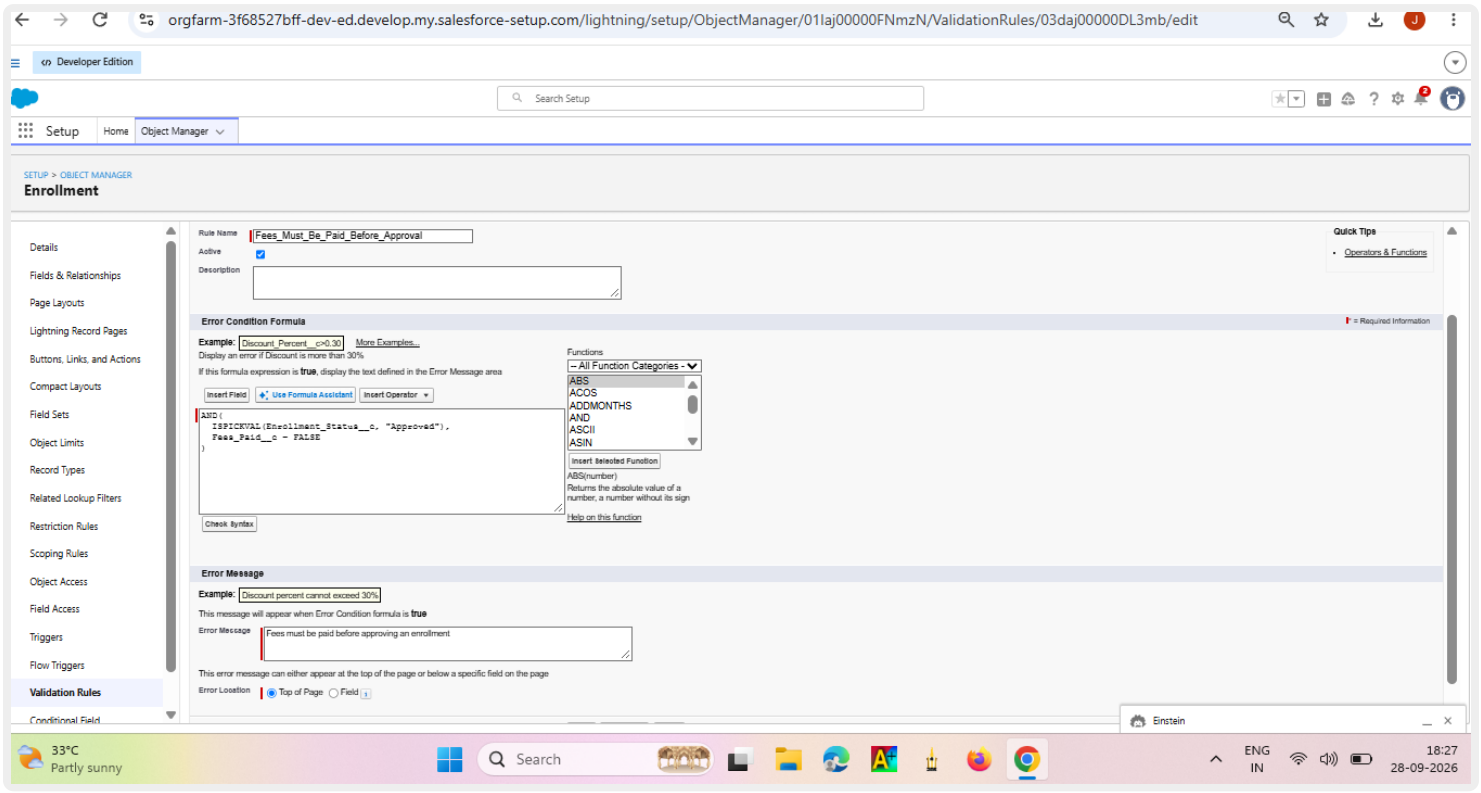
Screenshot 02 — Enrollment Object — Fields & Relationships (continued)



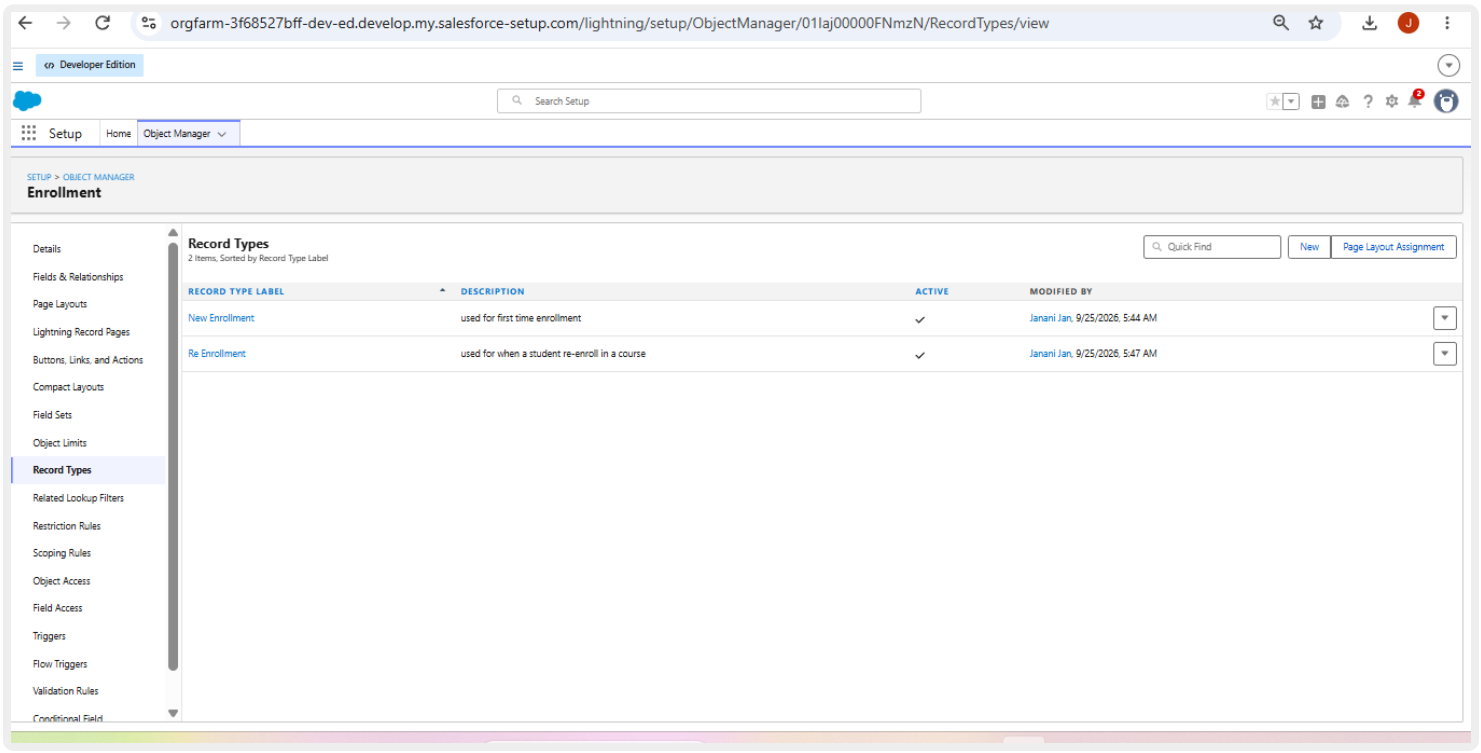
Screenshot 03 — Enrollment Object — Page Layouts



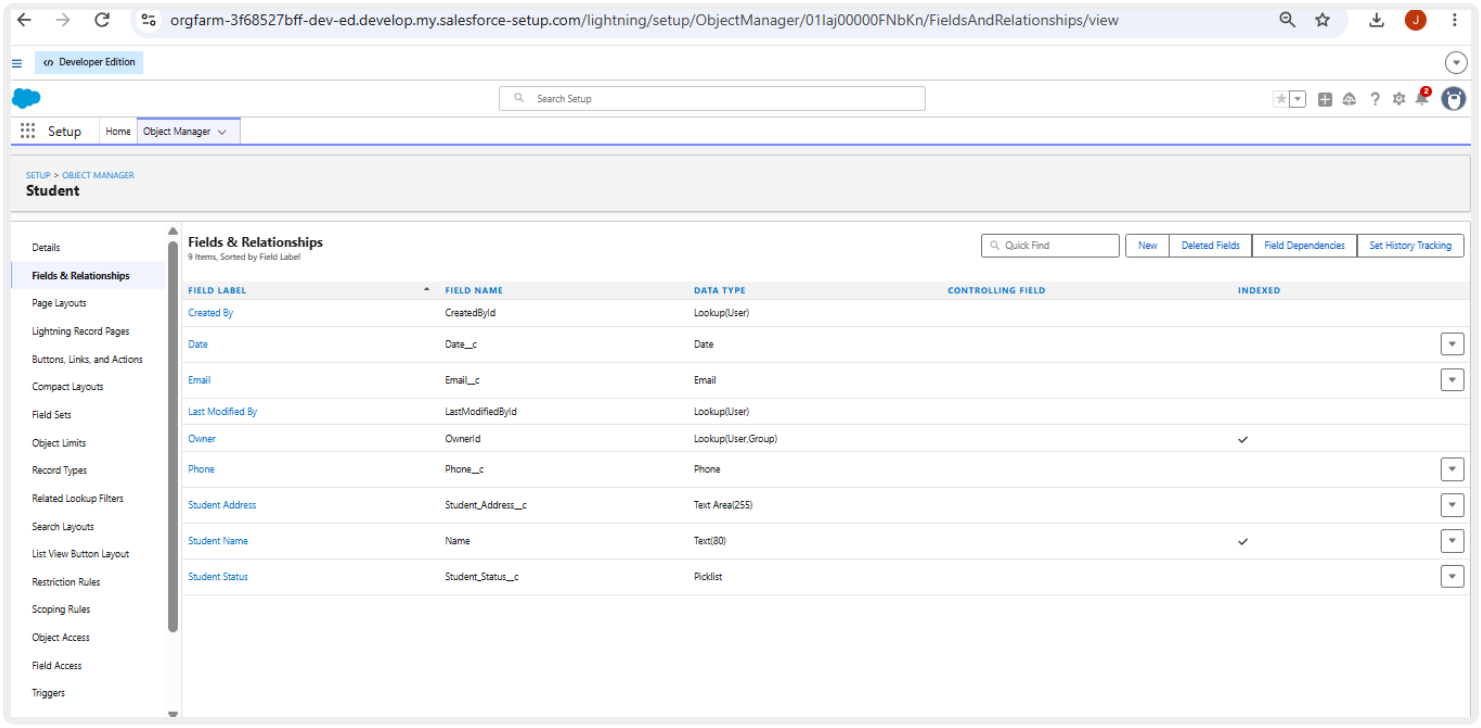
Screenshot 04 — Enrollment Validation Rule — Fees Must Be Paid Before Approval



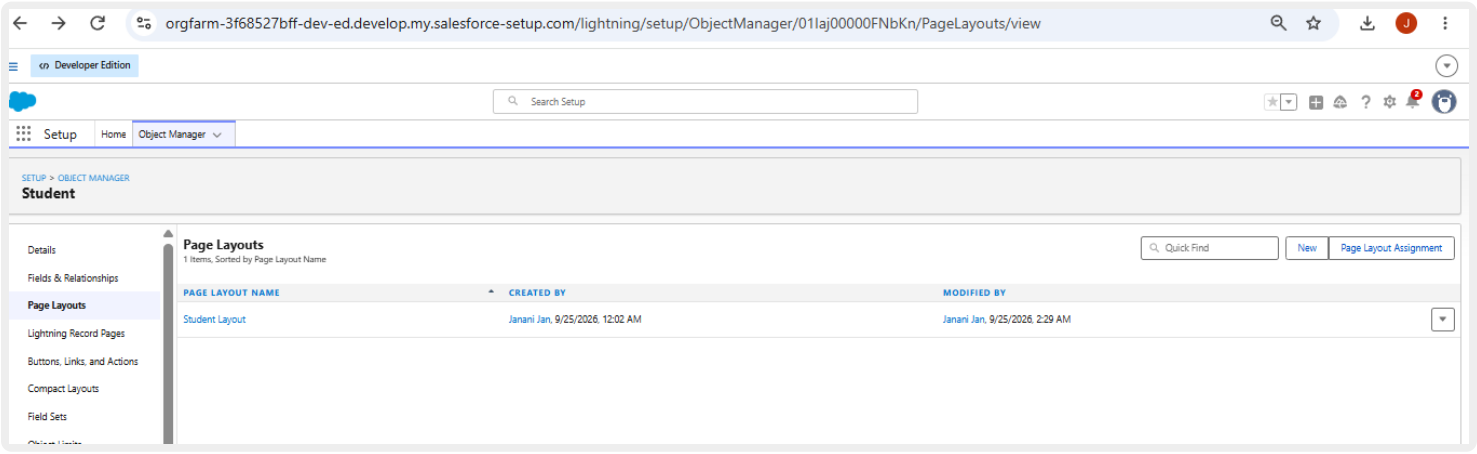
Screenshot 05 — Enrollment Object — Record Types



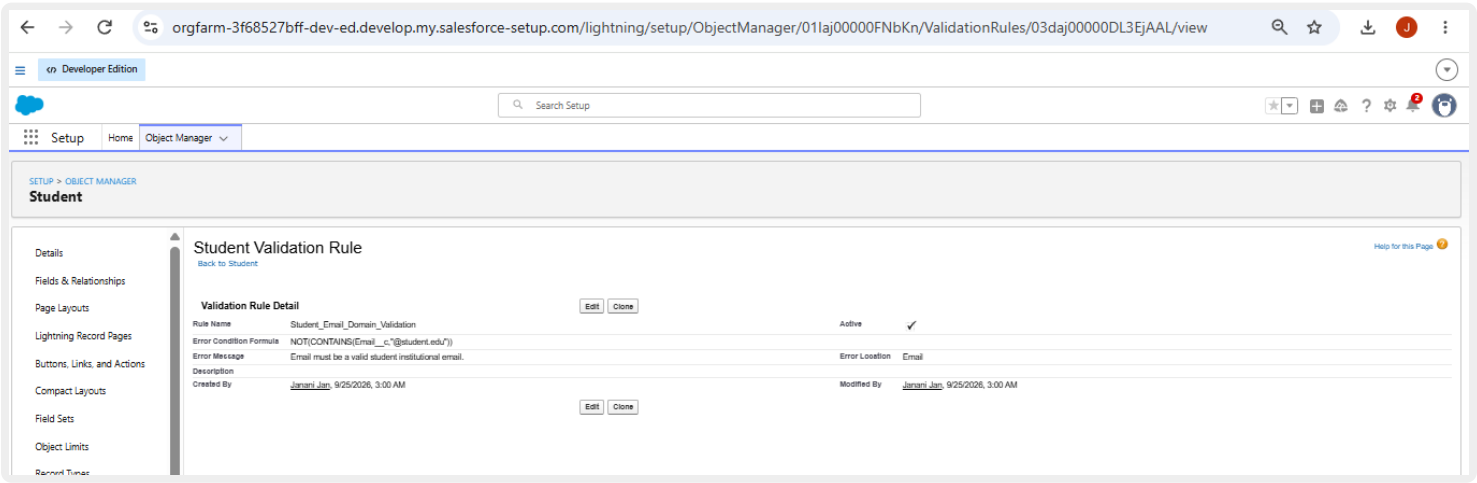
Screenshot 06 — Student Object — Fields & Relationships



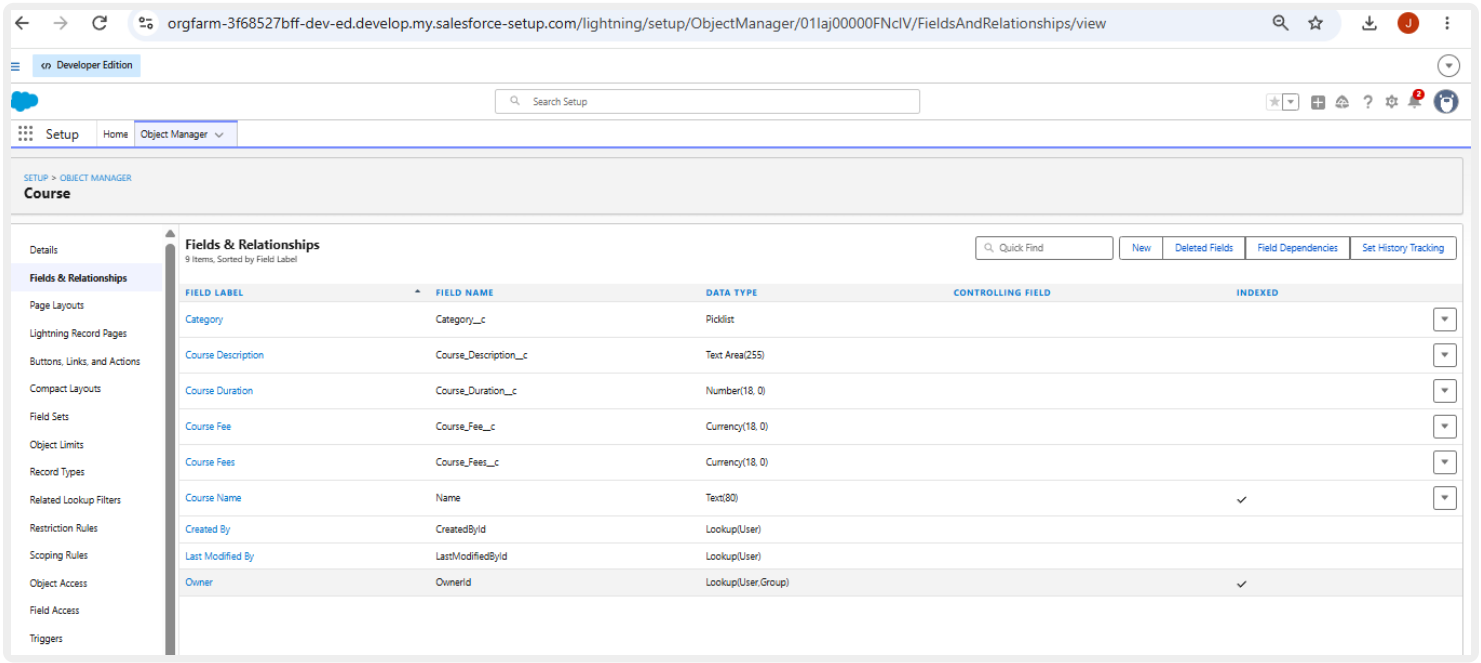
Screenshot 07 — Student Object — Page Layouts



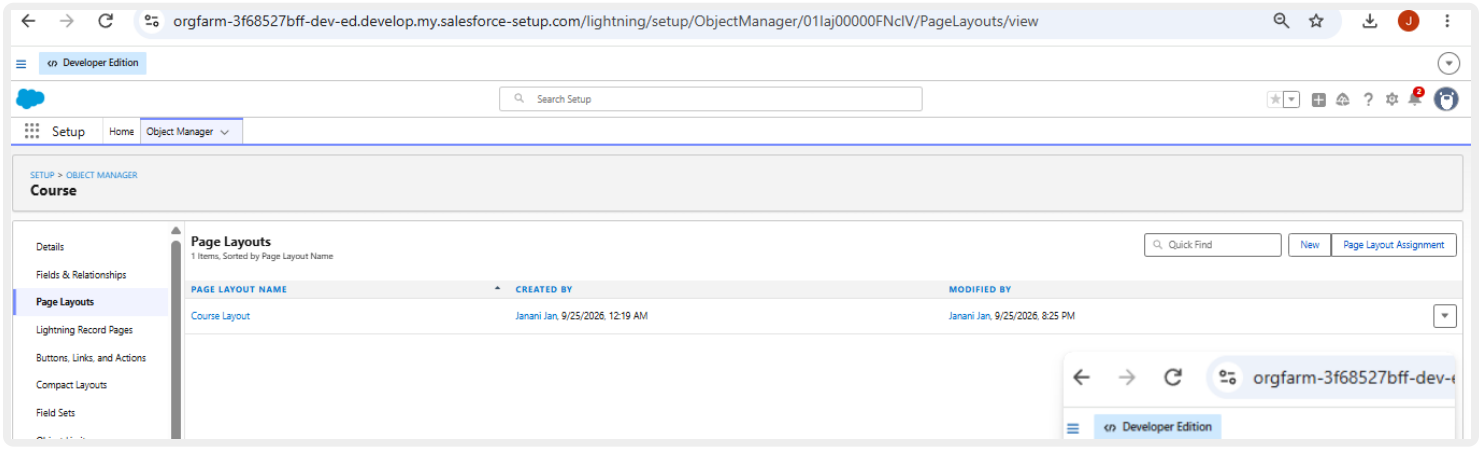
Screenshot 08 — Student Validation Rule — Student Email Domain Validation



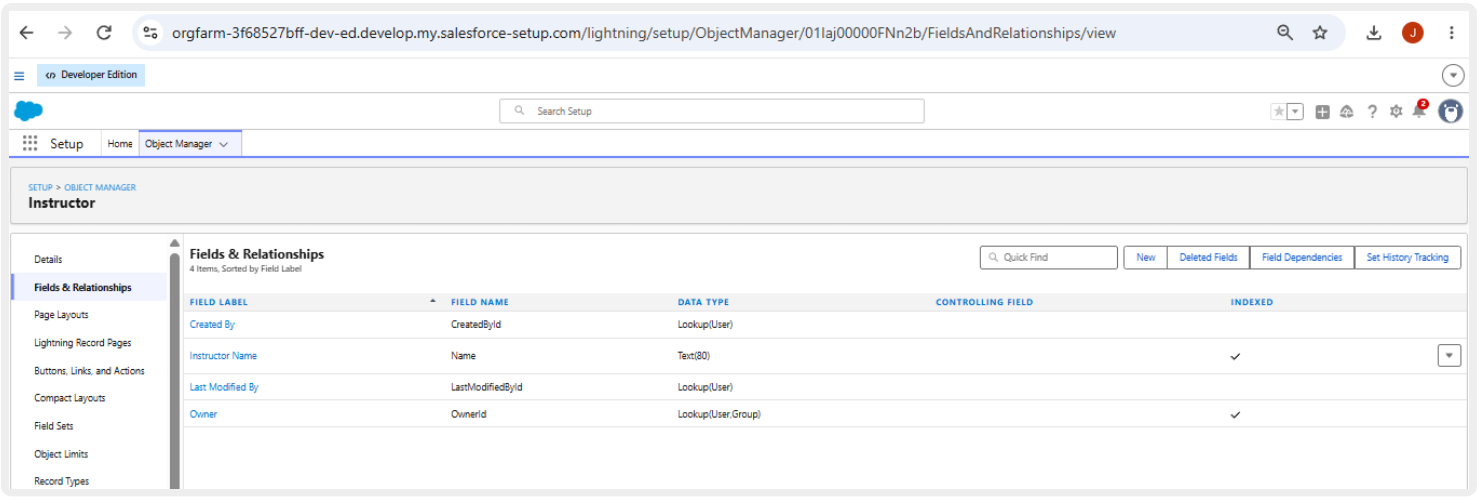
Screenshot 09 — Course Object — Fields & Relationships



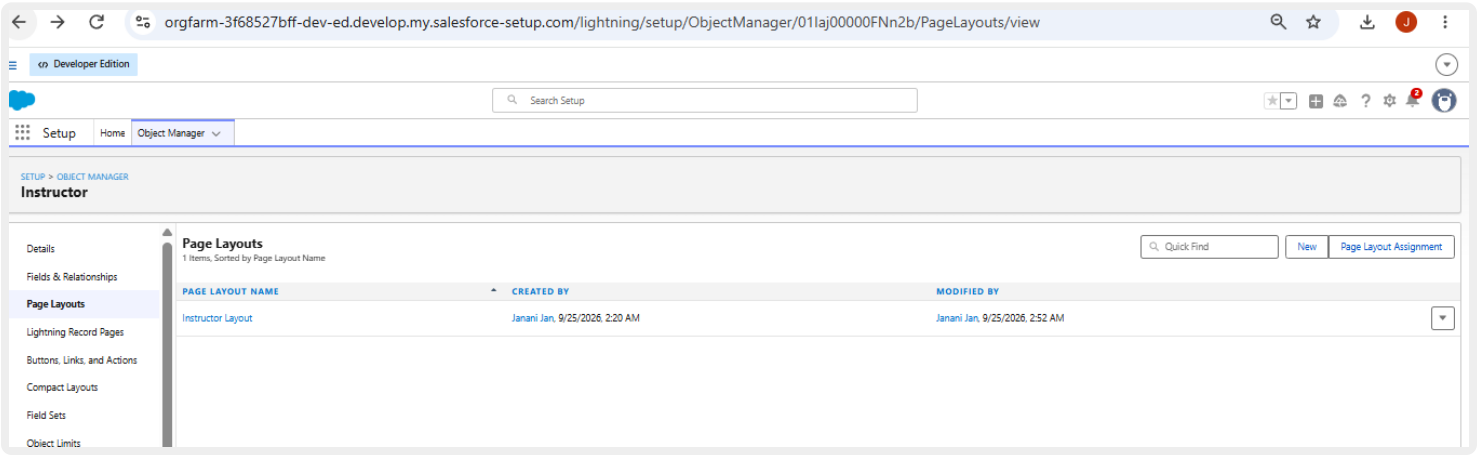
Screenshot 10 — Course Object — Page Layouts



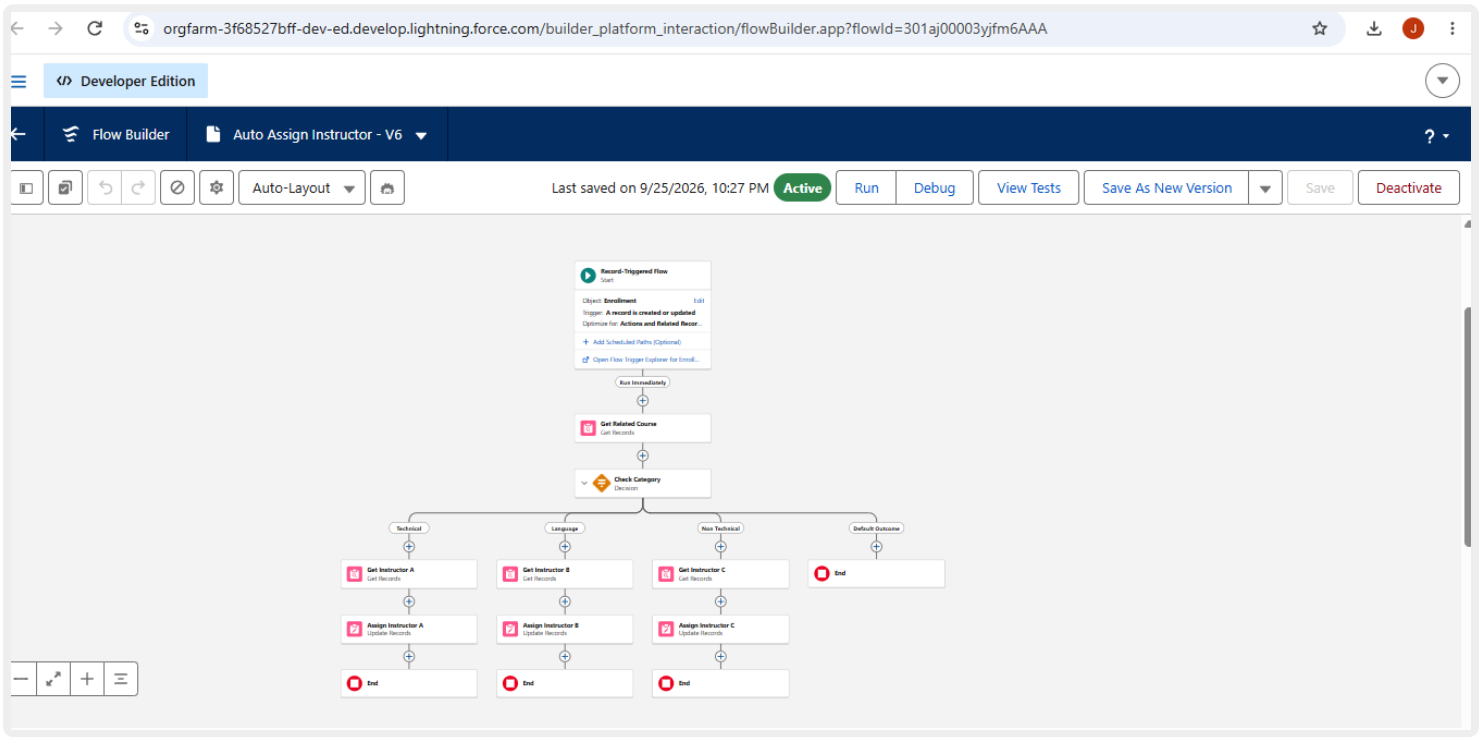
Screenshot 11 — Instructor Object — Fields & Relationships



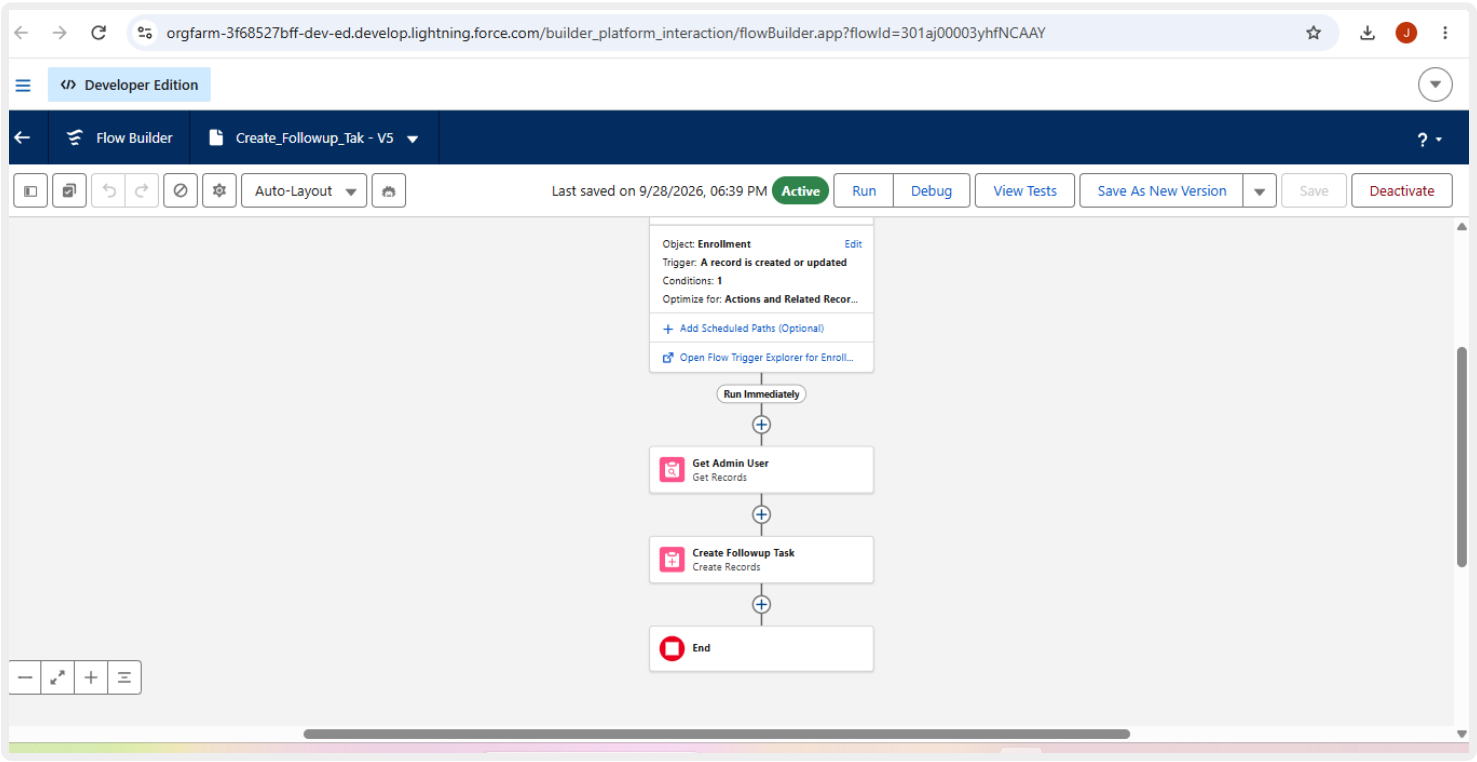
Screenshot 12 — Instructor Object — Page Layouts



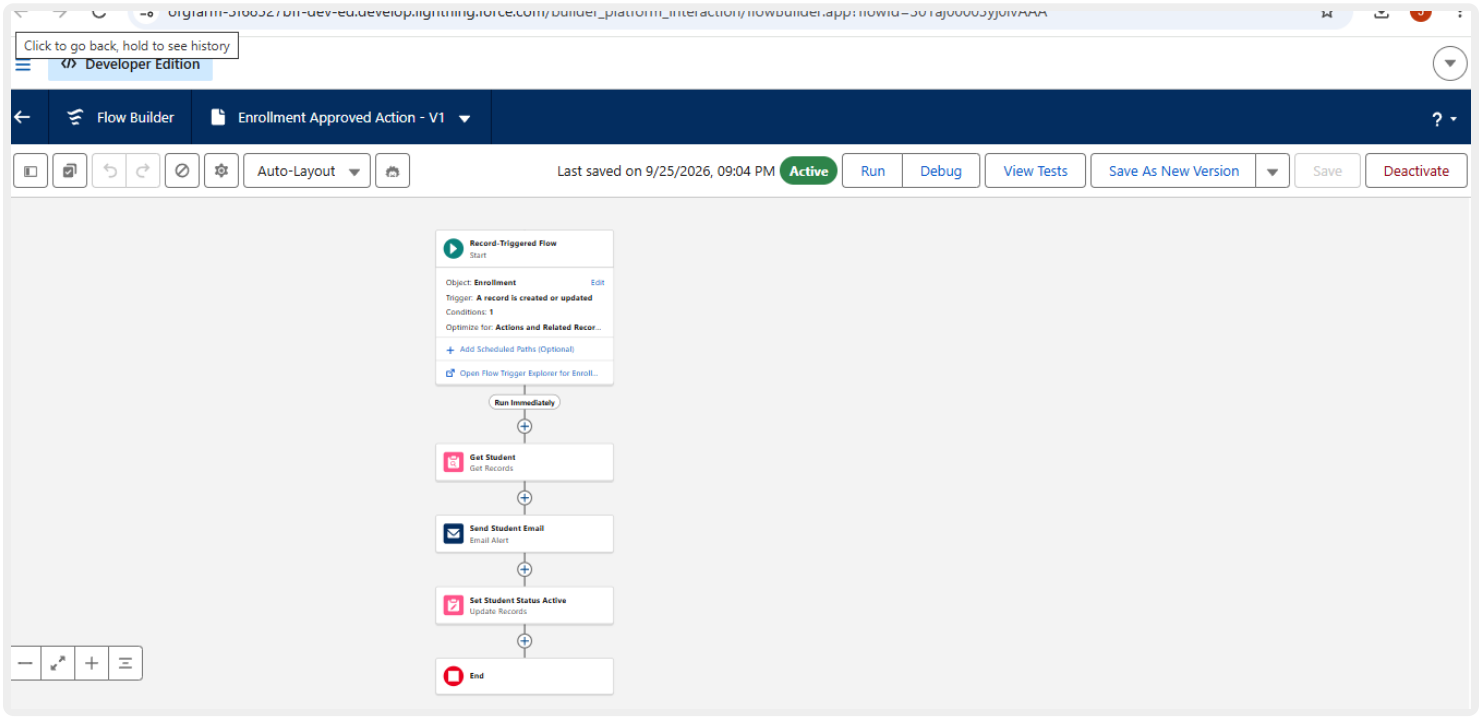
Screenshot 13 — Flow — Auto Assign Instructor



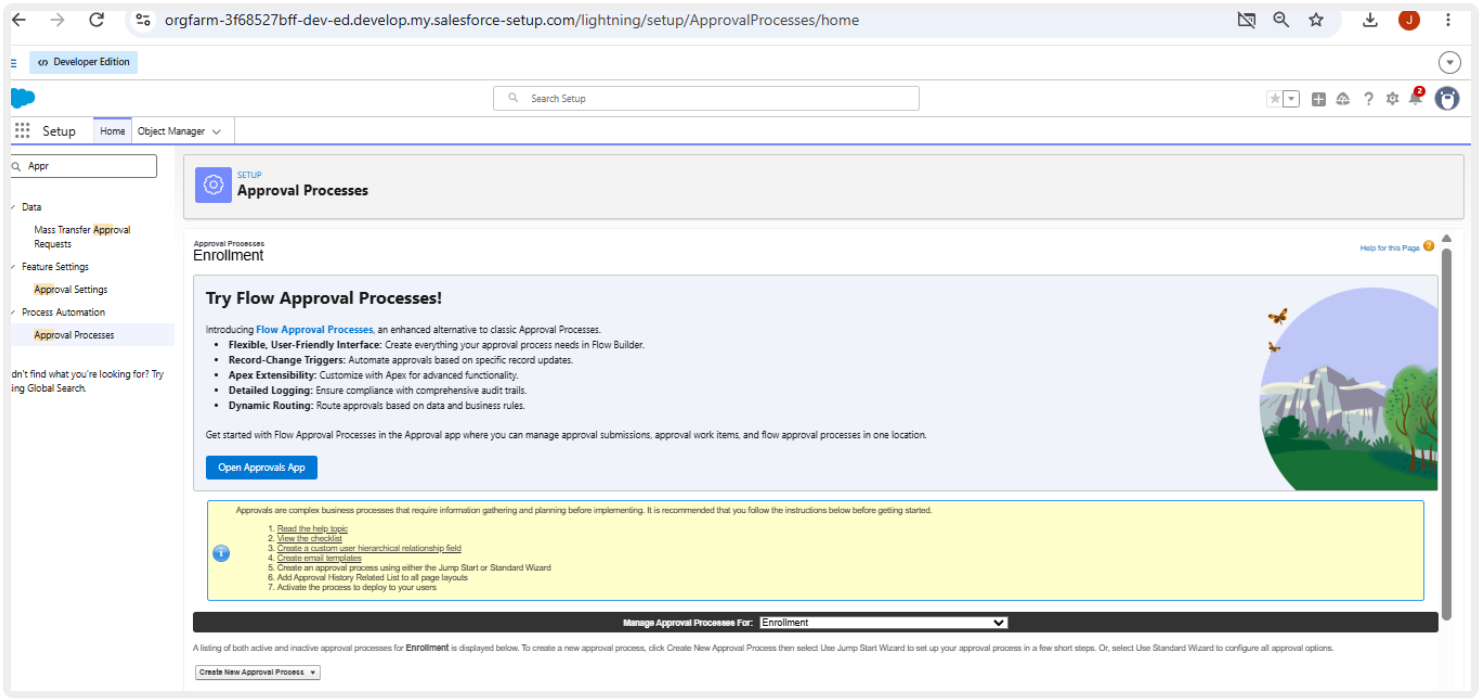
Screenshot 14 — Flow — Create Follow-up Task



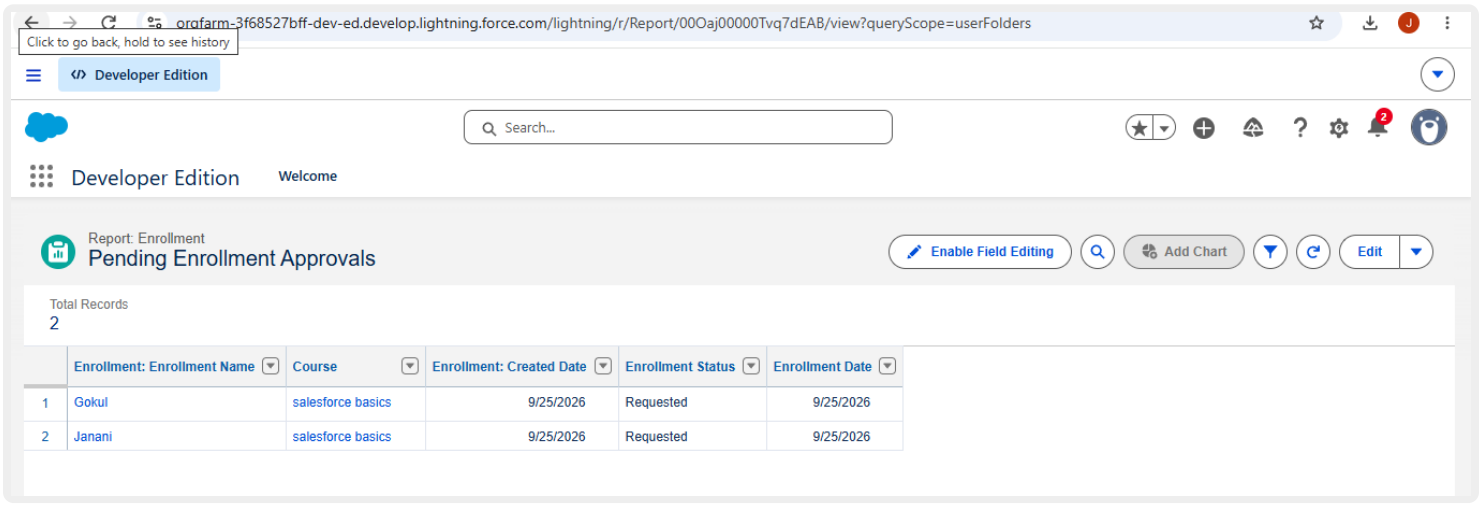
Screenshot 15 — Flow — Enrollment Approved Action



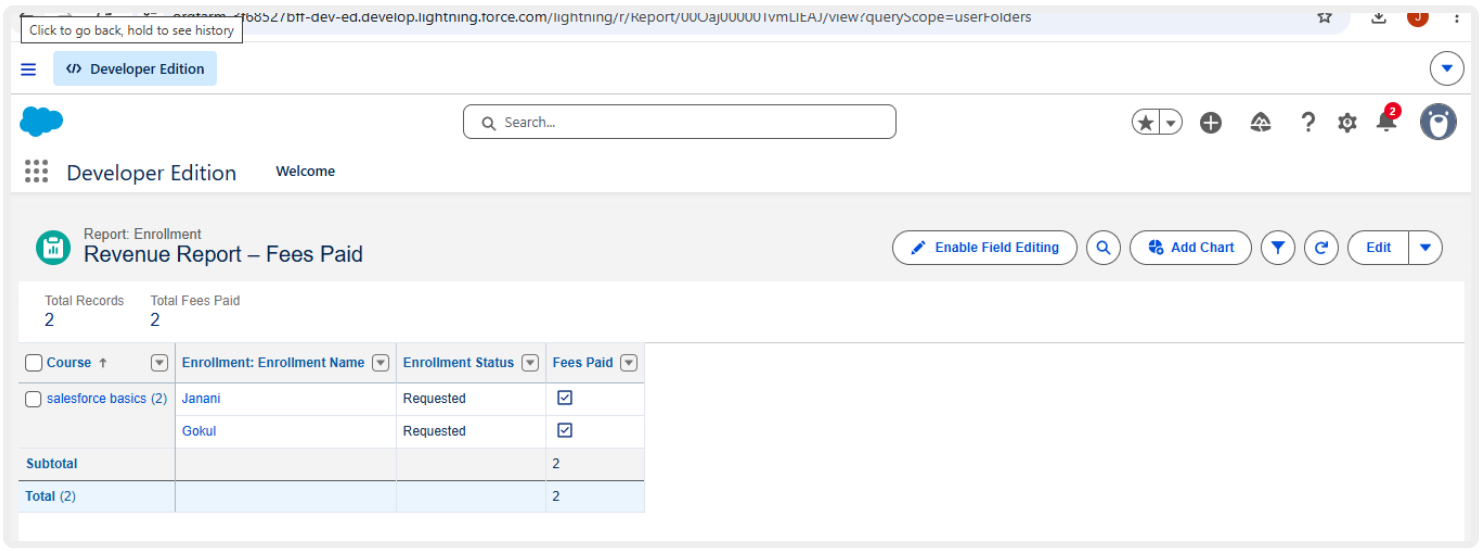
Screenshot 16 — Approval Processes — Enrollment



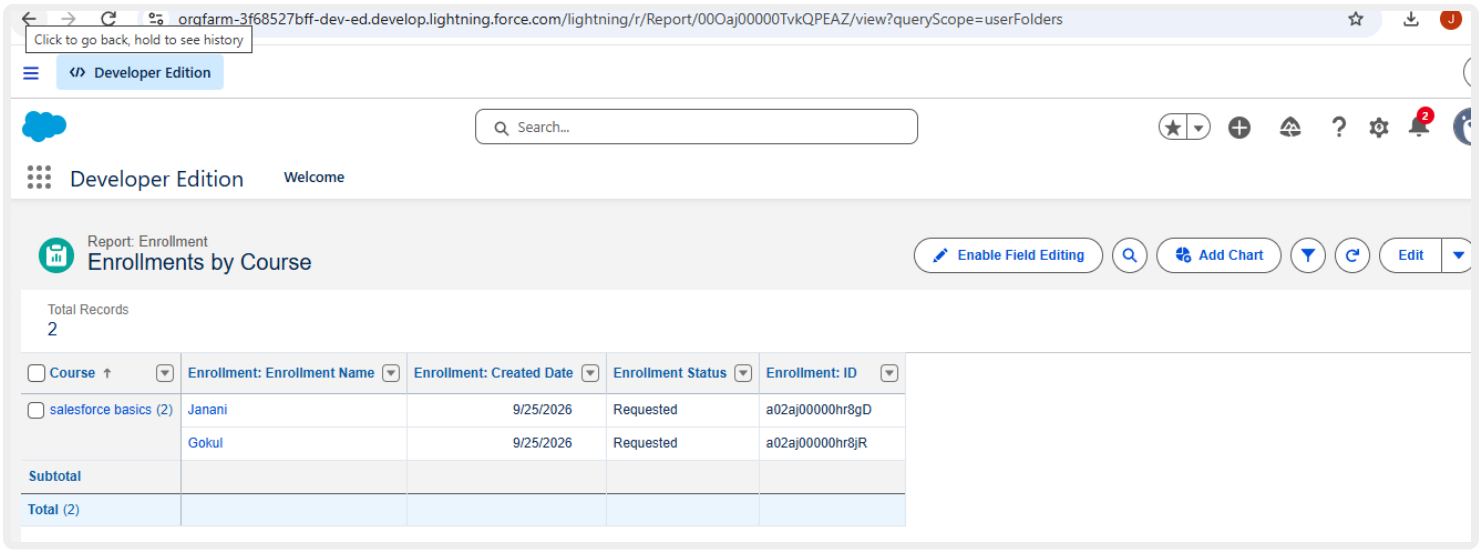
Screenshot 17 — Report — Pending Enrollment Approvals



Screenshot 18 — Report — Revenue Report / Fees Paid



Screenshot 19 — Report — Enrollments by Course



Screenshot 20 — Report — Students by Status

